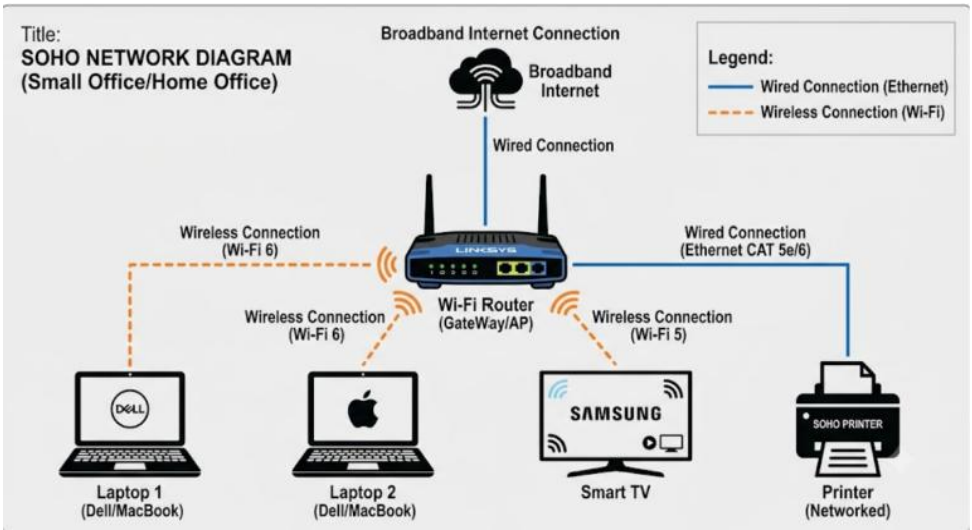
Build a SOHO Network

1. **Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).**



Device / Connection	**Type**	**Explanation**
Broadband Internet → Wi-Fi Router	**Wired**	Provides Internet to the router.
Laptop 1 → Wi-Fi Router	**Wireless (Wi-Fi)**	Laptop connects wirelessly to the network.
Laptop 2 → Wi-Fi Router	**Wireless (Wi-Fi)**	Laptop connects wirelessly to the network.
Smart TV → Wi-Fi Router	**Wireless (Wi-Fi)**	TV accesses the Internet through Wi-Fi.
Printer → Wi-Fi Router	**Wired (Ethernet)**	Printer connects using an Ethernet cable.

**2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.**

To build the SOHO network, the following hardware components will be required:

1. **Broadband Internet Connection / ONT Modem – 1**
   An ISP-provided modem or ONT, such as Airtel Xstream Fiber or JioFiber, will be used to connect the network to the Internet.

2. **Wi-Fi Router – 1**
   A router such as the TP-Link Archer C6 AC1200 can be used. It will provide Wi-Fi connectivity to the laptops, Smart TV, and printer.

3. **Laptops – 2**
   Two Wi-Fi-enabled laptops will be connected to the router wirelessly.

4. **Smart TV – 1**
   A Wi-Fi-enabled Smart TV from brands such as Samsung, LG, Sony, or Xiaomi can be connected to the router through Wi-Fi.

5. **Wi-Fi Printer – 1**
   A printer such as the Canon PIXMA E3370 can be used. It can connect to the network wirelessly and allow the laptops to print documents.

6. **CAT6 Ethernet Cable – 1–2**
   A D-Link CAT6 UTP Ethernet cable can be used to connect the broadband modem/ONT to the Wi-Fi router. One extra cable can be kept for backup or for connecting another device through Ethernet.

7. **Power Adapters – As Required**
   The power adapters supplied with the modem, router, printer, and other devices will be required to power them.

8. **Power Extension Board – 1**
   A power extension board can be used to provide electricity to the modem/ONT and router if there are not enough nearby power sockets.

3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

1. Connect the router: Connect the broadband cable to the router's WAN/Internet port and switch it on.

2. Connect a laptop: Connect a laptop to the router using Wi-Fi or a LAN cable.

3. Open router settings: Open a browser and enter the router's IP address, such as 192.168.1.1.

4. Login: Enter the default username and password given with the router.

5. Set SSID: Go to Wireless Settings and change the Wi-Fi name to a unique name

6. Enable WPA2: In Security Settings, select WPA2-Personal (WPA2-PSK) and set a strong Wi-Fi password.

7. Change admin password: Go to Administration/Management and replace the default admin password with a strong, unique password.

8. Save settings: Click Save/Apply and allow the router to restart.

9. Reconnect devices: Connect the two laptops, Smart TV, and printer using the new Wi-Fi name and password.

10. Test the network: Check that all devices can access the Internet and communicate with each other.

4. For this SOHO network, I would use the private IP range 192.168.1.0/24. The router will act as the DHCP server and automatically assign IP addresses to the two laptops. The printer and smart TV will have fixed/static IP addresses so that they can be easily found on the network.

Device	MAC Address	IP Address	IP Assignment
Wi-Fi Router	00:1A:2B:3C:4D:01	192.168.1.1	Static
Laptop 1	00:1A:2B:3C:4D:02	192.168.1.100	DHCP
Laptop 2	00:1A:2B:3C:4D:03	192.168.1.101	DHCP
Smart TV	00:1A:2B:3C:4D:04	192.168.1.10	Static
Printer	00:1A:2B:3C:4D:05	192.168.1.11	Static

5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

If one laptop cannot access the Internet, I would check the following common issues:

1. **Wi-Fi Connection Problem**
   Check whether Wi-Fi is turned on and the laptop is connected to the correct SSID. Reconnect using the correct password.
2. **IP Address/DHCP Problem**
   Use ipconfig to check whether the laptop has a valid IP address. If not, use ipconfig /renew to get a new IP from the router.
3. **Router/Internet Problem**
   Check the router's Internet/WAN light and test another device. Restart the router if needed. If all devices are offline, contact the ISP.
4. **DNS Problem**
   If Wi-Fi works but websites do not open, check DNS. Try using 8.8.8.8 as the DNS server.
5. **Network Adapter Problem**
   Restart the laptop's Wi-Fi adapter or update its network driver if the connection still does not work.